

1 True/False

1. iBGP is used for intradomain routing.

Solution: False, IGP, such as OSPF or RIP, is used for intradomain routing

2. Avoiding loops is one reason why BGP uses path vector.

Solution: True, also helps ASes adopt certain policies

3. BGP always advertises a shortest path.

Solution: False, the path relies on other policies (like money) instead of purely distances

4. BGP route advertisements use classless addressing.

Solution: True

2 Interdomain vs Intradomain

Given four ASes Verizon, Comcast, Berkeley and Stanford, with border routers V1, C1, B1 and S1 respectively, and internal routers V2, C2, B2 and S2 respectively, we know Berkeley and Stanford both use Comcast's and Verizon's services. And the (fake) cost metrics are 10/MB for using Comcast's bandwidth and 20/MB for using Verizon's bandwidth. Please answer following questions with the assumption made in lecture: a border router establishes iBGP sessions to all other routers within its AS.

1. Which one of eBGP, iBGP and IGP distributes externally learned routes internally, and which routers, if any, speak it?

Solution: iBGP. V1, C1, B1, S1, V2, C2, B2, S2. All routers speak iBGP.

2. Which one of eBGP, iBGP and IGP learn routes to external destinations, and which routers, if any, speak it?

Solution: eBGP. V1, C1, B1, S1. Only border routers speak eBGP.

3. Which one of eBGP, iBGP and IGP provides internal reachability, and which routers, if any, speak it?

Solution: IGP. V1, C1, B1, S1, V2, C2, B2, S2. All routers speak IGP

4. Which AS would Berkeley use to reach Stanford, in terms of cost effectiveness?

Solution: Comcast. It costs less amount of money.

5. Given now Comcast knows Berkeley and Stanford don't get along with each other, it doesn't advertise its route of Berkeley to Stanford, or the other way around. However, Verizon still remains neutral. Which AS would Berkeley use to reach Stanford now?

Solution: Verizon.

3 BGP

Consider the following configuration of Autonomous Systems and their relationships.

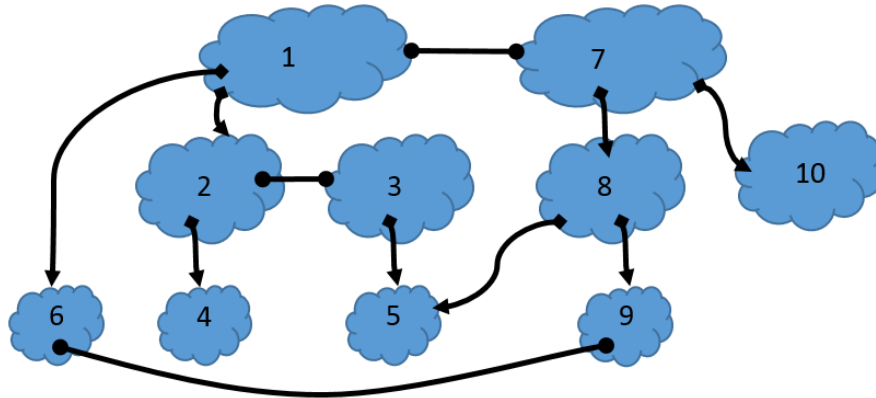


Figure 1: Autonomous System Relationships

A square-triangle line indicates a provider-customer relationship, and a circle-circle line indicates a peer relationship. We have two top-level (Tier 1) providers 1 and 7.

Assume we are using the following selection and export rules for all ASes:

Selection: Prefer to send data through customers > peers > providers in order to minimize costs.

Export: To customers, advertise all paths. To all other ASes, advertise only paths to customers (and paths to your own AS, of course).

We present the following questions.

1. A user in AS 4 tries to send data to a friend in AS 5. What path (if any) would the data take in order to get to the friend?

Solution: 4, 2, 3, 5

2. A user in AS 6 tries to send data to a friend in AS 5. What path (if any) would the data take in order to get to the friend?

Solution: 6, 1, 7, 8, 5. This is trickier since 2 won't advertise its peer's (3's) path (about how to reach 5) to its provider (1), so 1 doesn't know it can reach 5 through 2.

3. A user in AS 1 tries to send data to another user in AS 9. What path (if any) would the data take in order to get there?

Solution: 1, 7, 8, 9. Same reasoning as part 2.

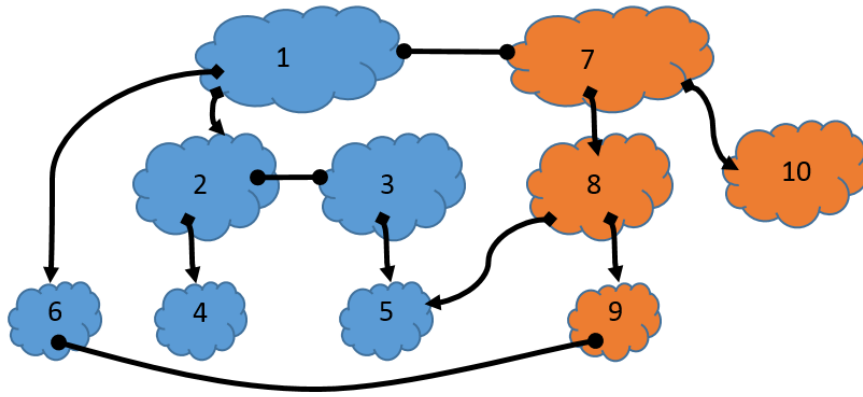


Figure 2: Affiliations

Now assume all Autonomous Systems are using the same selection policy but also adopting new export rules because they are part of nation states in an information war. Autonomous Systems ranging from 1 to 6 form a group of allies, and Autonomous Systems ranging from 7 to 10 form another group of allies:

Export:

1. To a customer AS that is an ally, advertise all paths.
2. To all other ASes that are allies, advertise only paths to customers.
3. Advertise nothing to ASes that are not allies.

We present the following questions with the new policies.

4. A user in AS 6 tries to send data to their friend in AS 5. Will the friend receive the data?

Solution: No. Similar to part 2, but this time, 7 is not advertising its path to 5.

5. Will a user in AS 9 be able to receive data from another in AS 1?

Solution: No. 7 advertises nothing to 1 because they are not allies.

6. Imagine AS 10 was a spy organization that was able to set up a deal with AS 1 to advertise all customer paths to AS 7. AS 10 now wants to report its findings to AS 5. Will the data be able to get from AS 10 to AS 5? If so, what path would the data take?

Solution: No. Even if AS 10 established this deal, 2 won't advertise to 1 about its path to 5, so 10 cannot talk to 5.